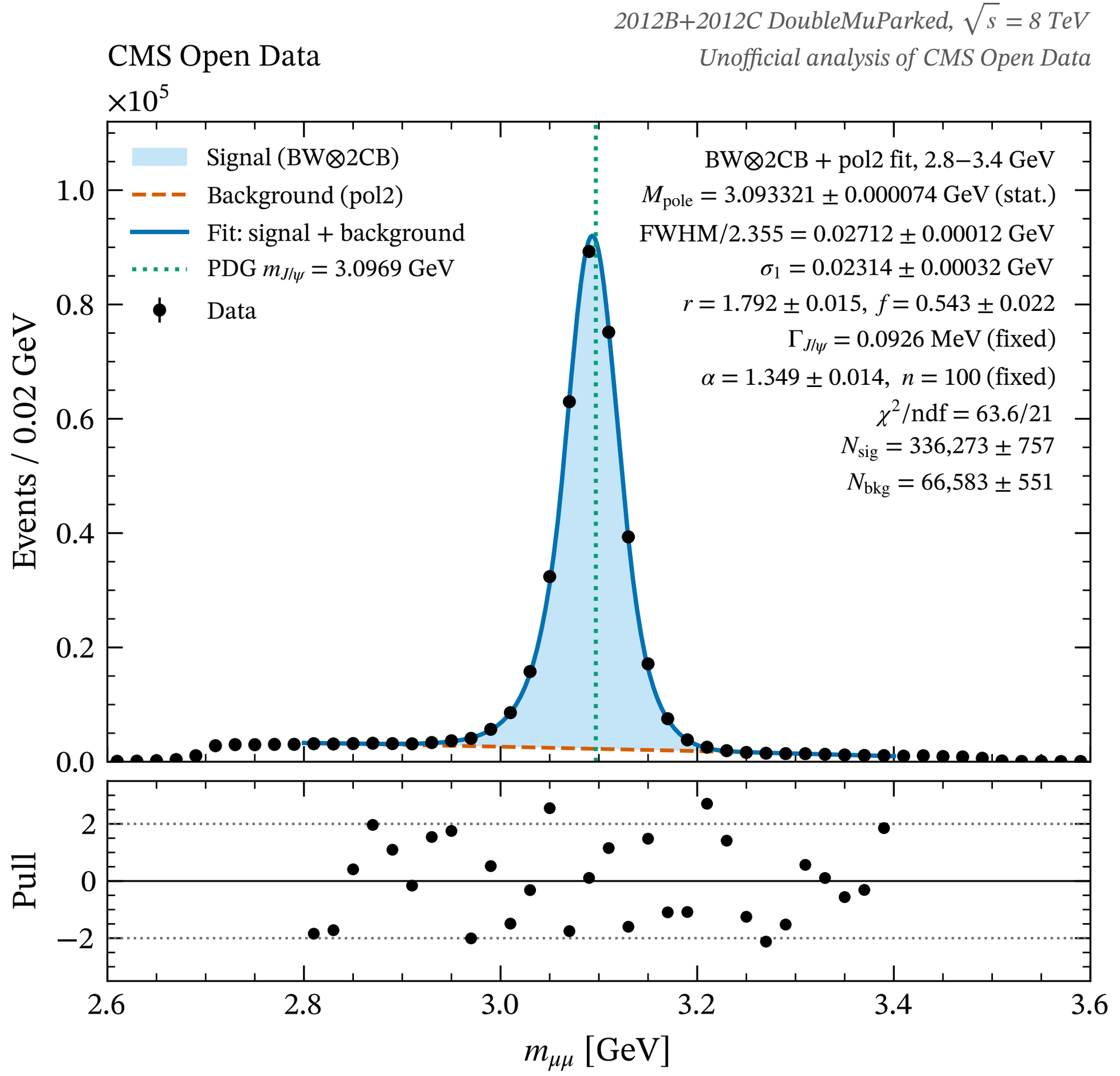

2012B+2012C DoubleMuParked, $\sqrt{s} = 8$ TeV
CMS Open Data
Unofficial analysis of CMS Open Data
$\times 10^5$
Signal (BW⊗2CB)
Background (pol2)
Fit: signal + background
PDG $m_{J/\psi} = 3.0969$ GeV
Data
BW⊗2CB + pol2 fit, 2.8−3.4 GeV
$M_{\text{pole}} = 3.093321 \pm 0.000074$ GeV (stat.)
FWHM/2.355 $= 0.02712 \pm 0.00012$ GeV
$\sigma_1 = 0.02314 \pm 0.00032$ GeV
$r = 1.792 \pm 0.015$, $f = 0.543 \pm 0.022$
$\Gamma_{J/\psi} = 0.0926$ MeV (fixed)
$\alpha = 1.349 \pm 0.014$, $n = 100$ (fixed)
$\chi^2/\text{ndf} = 63.6/21$
$N_{\text{sig}} = 336{,}273 \pm 757$
$N_{\text{bkg}} = 66{,}583 \pm 551$
Events / 0.02 GeV
1.0
0.8
0.6
0.4
0.2
0.0
Pull
2
0
−2
2.6
2.8
3.0
3.2
3.4
3.6
$m_{\mu\mu}$ [GeV]